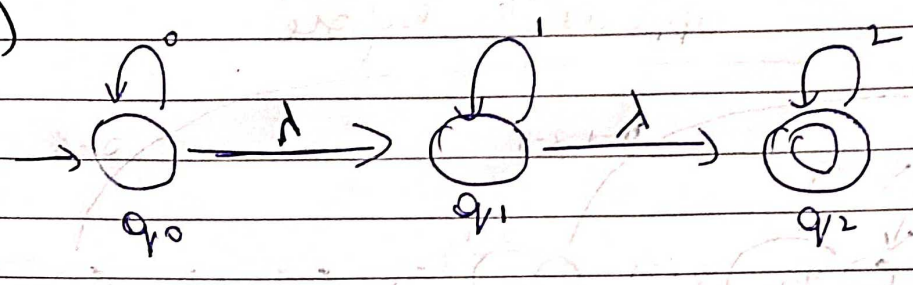
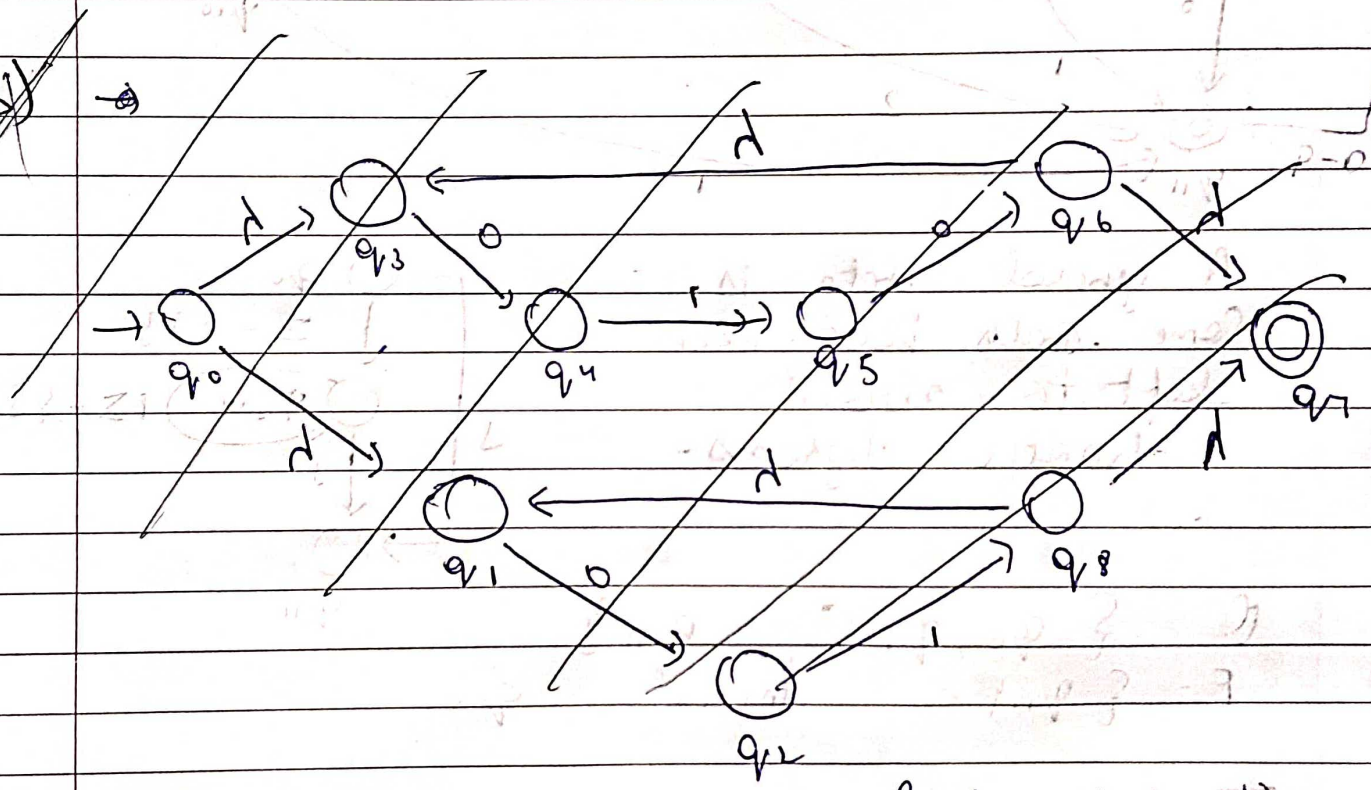
Assignment - 3

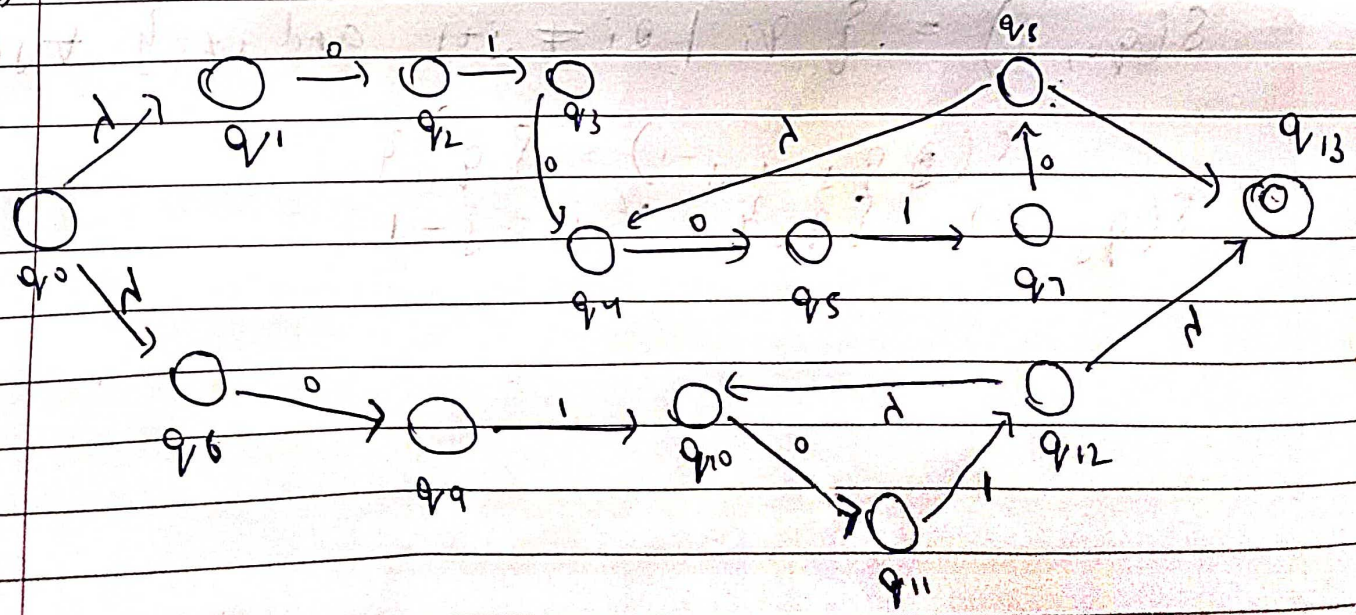
Q1) (a)



~~(b)~~



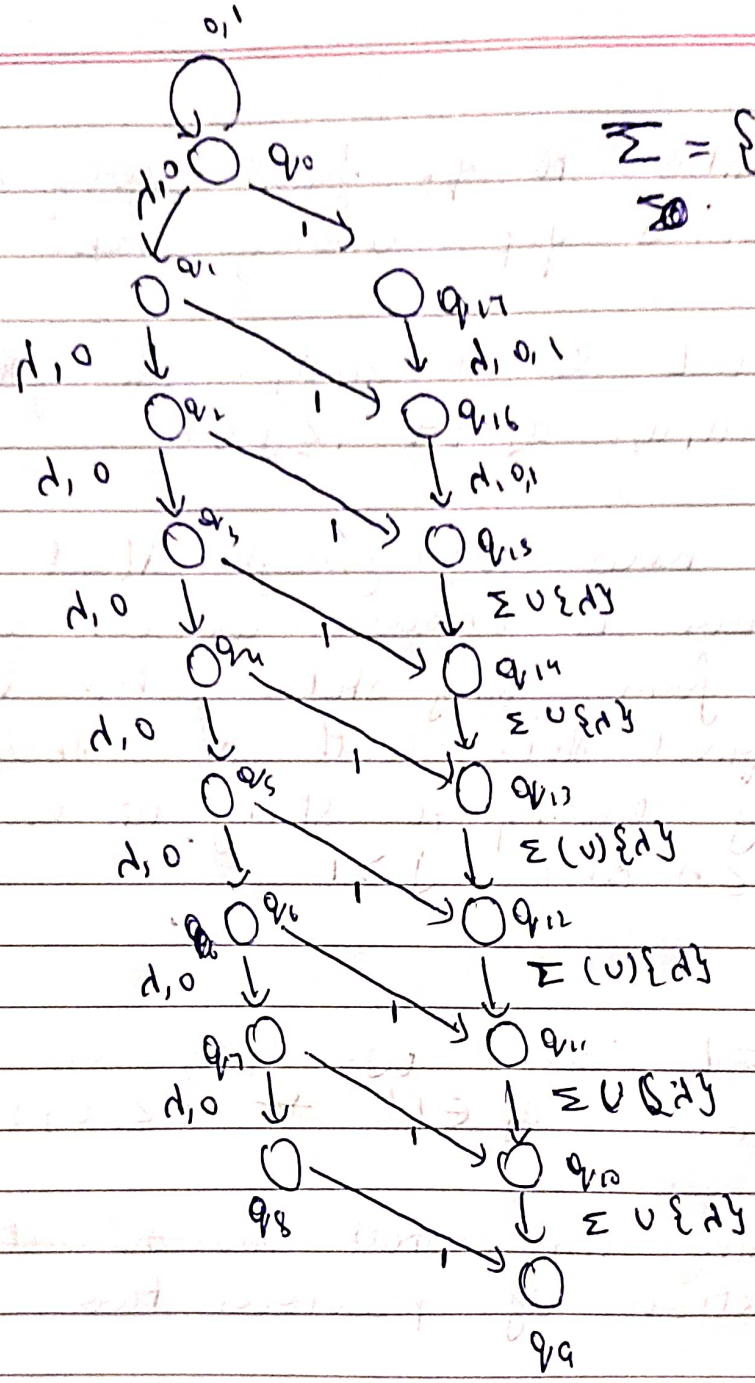
(b) if repetition means at least two times



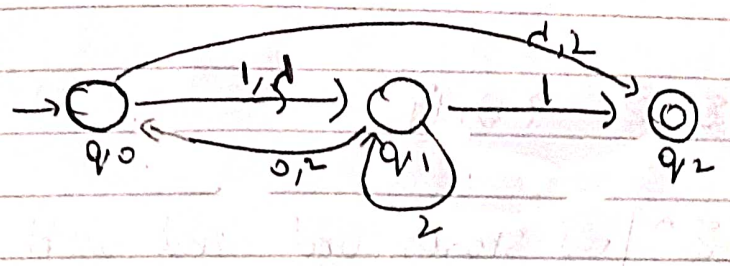
Page No.:
Date:

9(c)

$\Sigma = \{0, 1\}$



Q2)



(a) Closure of d-NFA states

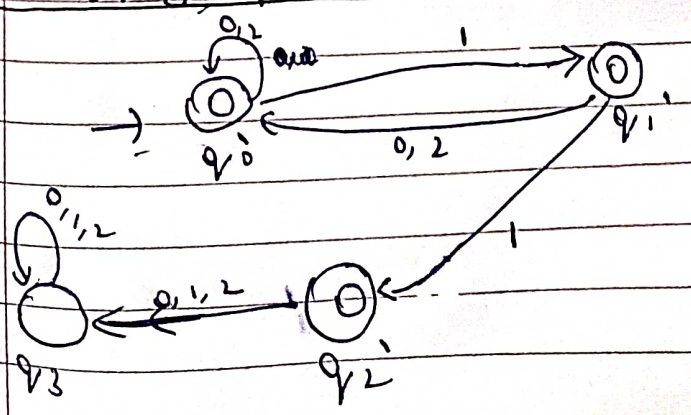
$d-cl(q_0) = \{q_0, q_1, q_2\}$
$d-cl(q_1) = \{q_1\}$
$d-cl(q_2) = \{q_2\}$

(b) Strings of length ≤ 3 that are accepted by the d-NFA are:-

=) { d, 2, 1, 21, 221, 11, 121, 000, 0, 00, 22, 222, 02, 002, 10, 102, 20, 202, 101, 211, 201, 011, 200, 122, 220, 001, 010, 012, 020, 021, 022, 100, 120, 210, 212, 01, 12

(c) Transition table using closures

	0	1	2
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_2, q_1, q_0\}$
$\{q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_2\}$	$\{q_2, q_0, q_1\}$
$\{q_2\}$	ϕ	ϕ	ϕ



$q_0' = \{q_0, q_1, q_2\}$
$q_1' = \{q_1, q_2\}$
$q_2' = \{q_2\}$
$q_3 \rightarrow$ Dead state

Page No.:
Date:

~~Transition~~

$\Sigma = \{0, 1\}$

Q4) (a)

$L = \{ w \in \Sigma^* \mid w$ starts and end with the same symbol

$re = 0(1+0)^*0 + 1(0+1)^*1$

(b) $((0+1)^* 11 (0+1)^* 010 (0+1)^*) + ((0+1)^* 010 (0+1)^* 11 (0+1)^*)$

(c) ~~$(0^* 1 0^* (10^* 1 0^*)^*) \cdot (1^* (01^* 01^*)^*) + (1^* (01^* 01^*)^*) \cdot (0^* 1 0^* (10^* 1 0^*)^*)$~~

(d) $(1+01)^* (00 \quad)(1+10)^*$

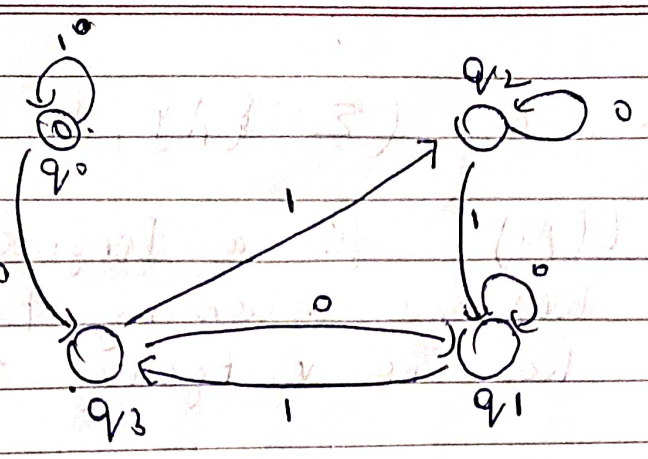
(e) $re = ((0+1)(0+1)(0+1))^*$

(f) $((00+01(11)^*10)^*(1+01(11)^*0)(0(11)^*0)^*(1+0(11)^*10))^* \ldots$

continued $\rightarrow$ $(00+01(11)^*10)^*(1+01(11)^*0)(0(11)^*0)^*$

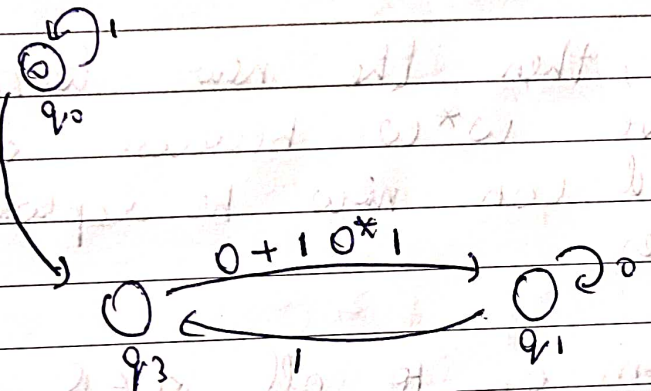
Page No :
Date:

Q5) →

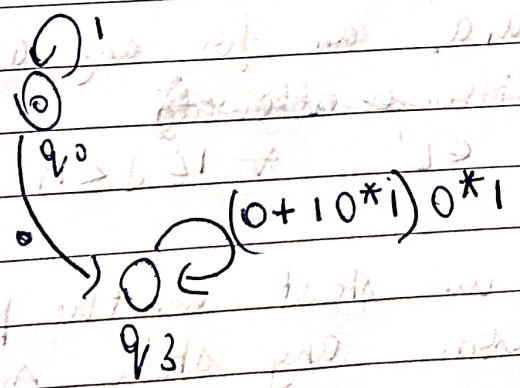


Eliminate q_2

In this question, q_3, q_1, q_2 turn out to be dead states



Eliminate q_1



Eliminate q_3

$= \boxed{1^*}$

Page No.:
Date:

Q3) Given : $N = (\Sigma \cup \{\lambda\}, Q, \delta, q_0, \{q_f\})$

$L = L(N)$ for a language L

→ q_0 has no incoming transitions

→ q_f has no outgoing transition

(a) λ- transition from q_f to q_0

If any $w \in L$, then the new language L' will have $w^* w$ because any accepted word can now be repeated zero or more times.

(b) λ- transition from q_0 to all $q \in Q$ st. q is reachable from q_0 along a path with labels $\Sigma \cup \{\lambda\}$.

for $w \in L$ st. $w = a_1 a_2 \ldots a_n$ for any $a_i \in \Sigma \cup \{\lambda\}$

then $a_j a_{j+1} \ldots a_n \in L'$, $\forall\ 1 \le j \le n$

This means that we can start the path to get a string from any state since there is a lambda transition till that state.*

L' → new language accepted by λ- NFA

* → that state must have a path from q_0.

Page No :
Date:

(c) λ transition to q_f from every state that reaches q_f along some path.

If $w \in L$ s.t $w = a_1 a_2 \cdots a_n$ for $a_i \in \Sigma \cup \{\lambda\}$, then $a_1 a_2 \cdots a_j \in L \; \forall \; 1 \leq j \leq n$

~~The~~ L' is new language that is accepted. The above is because we can go to ~~any~~ the final state from any state which has a path to the final state with a lambda transition thereby ending the string at any letter a_j s.t $j \leq n$ and $j \geq 1$

(d) If $w \in L$ s.t $w = a_1 a_2 \cdots a_n$ then $a_i a_{i+1} \cdots a_j \in L' \; \forall \; 1 \leq i, j \leq n$ and $i \leq j$

~~These string can start and end at any~~
Combination of previous two effects.